

MMS Insights into CME Driven Sub-Alfvénic Solar Wind at 1 AU

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ABSTRACT

We report the properties of electron distributions and turbulence during a Coronal Mass Ejection (CME) in April 2023 observed by Magnetospheric Multiscale (MMS). The CME exhibits a clear sheath and magnetic cloud (MC), and within the MC, the solar wind becomes sub-Alfvénic for two hours. We investigate plasma and turbulence properties of the sub-Alfvénic CME wind and compare them with those in the super-Alfvénic solar wind in the MC and CME sheath. Electrons within the sub-Alfvénic MC show significantly higher temperatures than those in the CME sheath and the super-Alfvénic MC, with their one-dimensional distributions revealing super-thermal tail and a depletion in electron populations between 15–50 eV. Within the CME sheath, isolated regions of electron heating are observed, where parallel energy flux is enhanced up to ~ 1 keV. Magnetic field fluctuations within the sub-Alfvénic MC interval exhibit negligible cross helicity and steeper-than-Kolmogorov scaling in the inertial range, with no clear spectral break. These fluctuations also show reduced intermittency at ion and sub-ion scales, emerging intermittency at electron scales, and weak magnetic compressibility. Together, these observations point to the presence of weak magnetohydrodynamic (MHD) turbulence within the sub-Alfvénic MC, resembling conditions commonly observed in planetary magnetospheres such as Jupiter’s.

1. INTRODUCTION

Coronal Mass Ejections (CMEs) are macro-scale magnetized structures containing plasma and magnetic field expelled from the Sun and are one of the key drivers of the space weather at the Earth (D. F. Webb & T. A. Howard 2012; J. T. Gosling 1993). CMEs can travel faster than the ambient solar wind driving an interplanetary (IP) shock ahead of them, that includes the shock front and the intervening CME sheath region. CMEs are commonly accompanied by a magnetic cloud (MC) structure with 1) enhanced magnetic field, 2) smooth rotation of the magnetic field direction and depressed proton temperature and plasma beta (L. F. Burlaga et al. 1981; L. W. Klein & L. F. Burlaga 1982; E. K. J. Kilpua et al. 2017). Typically the plasma within the CME MC expands as it passes, and when it reaches the Earth, the conditions within are super-Alfvénic $M_A \sim 5$, where M_A is the Alfvén mach number defined as the ratio of bulk ion flow to the Alfvén speed. Under very rare conditions the

solar wind inside the MC are sub-Alfvénic ($M_A < 1$), mostly occur on average once every 2 years (E. Chané et al. 2012; R. Hajra & B. T. Tsurutani 2022) and are often associated with extreme solar wind conditions within CMEs (B. Lavraud & J. E. Borovsky 2008). These conditions inside MC include enhanced interplanetary magnetic field (IMF) and typically low solar wind plasma density, both of which increase the Alfvén speed and decrease M_A (C. F. Bowers et al. 2025).

One such rare occurrence took place on April 23rd 2023, a CME flux rope erupted on the surface of the Sun, driving a magnetic cloud. On April 24th 2023, the CME-MC passed over the Magnetospheric Multiscale (MMS) (J. L. Burch et al. 2016) spacecraft which were positioned at the dawn flank of the magnetopause ($X_{\text{GSE}} = [10.7, -7.9, -6.8]R_E$, in Geocentric Solar Ecliptic (GSE) coordinates) during its inbound trajectory. Fig. 1 provides an overview of the 2023 CME event, the IP shock driven by the CME arrives around 17:00 UT and MMS sees the CME sheath region from 17:40 to 01:10 UT. The figures show data from Fast Plasma Investigation (FPI) (C. Pollock et al. 2016) and

Fluxgate Magnetometer (FGM) (C. T. Russell et al. 2016) instruments, which provided observations of plasma properties and magnetic field conditions from the MMS spacecraft. From 02:00 UT to about 12:40 UT MMS observes the shocked solar wind as marked in the figure. MMS encountered magnetic cloud in two different intervals labeled as MCi-I and MCi-II in Fig. 1, each characterized by different solar wind conditions. MCi-I corresponds to the leading edge of the MC, where the solar wind remained super-Alfvénic, while MCi-II represents the MC’s interior, where sub-Alfvénic solar wind conditions prevailed. The magnetic fields within these two MC intervals, along with the smooth rotation in the magnetic field following MCi-I, resemble the MC interval observed by the Wind spacecraft (L.-J. Chen et al. 2024), which supports the identification of the two MCs. The sub-Alfvénic ($M_A \sim 0.6$) solar wind conditions within MCi-II was confirmed by measurements from OMNI spacecraft (shown in Fig. 1e) and is caused by the low density and high IMF strength in the MC rather than a decrease in the solar wind speed (L.-J. Chen et al. 2024). MMS observed sub-Alfvénic solar wind conditions for the first time, lasting approximately two hours, during which it recorded one hour of steady sub-Alfvénic flow followed by a 40-minute interval in which the typical unified structure of Earth’s magnetosphere—comprising the bow shock and magnetotail was split into Alfvén wings (L.-J. Chen et al. 2024). The magnetosphere transforms into of Alfvén wings due to the lack of a bow shock, a direct consequence of the sub-Alfvénic flow (L.-J. Chen et al. 2024) and recent work have shown the formation of Alfvén wings using global magnetohydrodynamic (MHD) simulations and have been confirmed by the MMS observations (H. Gurram et al. 2025; B. L. Burkholder et al. 2024; Y. Chen et al. 2025). A study by M. R. Argall et al. (2025) have studied turbulence inside the CME sheath and how turbulence evolves from Lagrange point (L1) to Earth using the Wind and MMS spacecraft. These studies have enabled the first-ever exploration of such phenomena at Earth, made possible by unprecedented measurement capabilities by MMS.

In this work, we present the first detailed analysis of steady sub-Alfvénic solar wind properties at 1 AU utilizing observations from the MMS. We performed a comparative analysis between two MC intervals i.e. sub-Alfvénic and super-Alfvénic MCs, with particular emphasis on the plasma characteristics and turbulence within these magnetically dominated structures originating from the solar corona. While the Parker Solar Probe has previously provided the earliest obser-

vations of sub-Alfvénic solar wind but that was within the Alfvénic surface near the Sun (R. Bandyopadhyay et al. 2022; R. Hajra & B. T. Tsurutani 2022; Y. Jiao et al. 2023), we will discuss the insights MMS dataset offers into the plasma properties and turbulence of sub-Alfvénic solar wind plasma at 1 AU. These observations help bridge knowledge across planetary systems, revealing how sub-Alfvénic flows shape the plasma environments both local and distant bodies—from Mercury and the Galilean moons to exoplanets orbiting close to their stars.

2. SOLAR WIND PROPERTIES INSIDE CME-DRIVEN MAGNETIC CLOUDS AND SHEATH

In this section, we analyze the solar wind properties observed inside a CME-driven MCs, focusing on two distinct intervals. During the first MC interval, the solar wind exhibited super-Alfvénic (MCi-I) conditions, whereas during the second interval, the solar wind was sub-Alfvénic (MCi-II) and we compare the plasma properties inside these two MCs: 1) MCi-I from 01:50 UT to 02:29 UT and 2) MCi-II from 12:30 UT to 14:04 UT. The key plasma parameters for both intervals are summarized in Table 1. In MCi-II, the proton number density was approximately 0.5 cm^{-3} , and the Alfvén Mach number was ~ 0.6 , and contrastingly MCi-I comprised of solar wind plasma with $n_e \sim 5$ and $M_A \sim 2$. The MCi-I was observed prior to the interplanetary magnetic field B_z reversal, with $B_z < 0$ throughout MCi-I, while MCi-II exhibited $B_z > 0$. Additionally, MCi-II featured a stronger magnetic field magnitude in comparison to MCi-I. The plasma beta, β , within MCi-II was nearly an order of magnitude lower than that in MCi-I, indicating a more magnetically dominated plasma. Furthermore, both electron and ion temperatures were elevated in the sub-Alfvénic MCi-II compared to the super-Alfvénic MCi-I, underscoring the distinct thermodynamic conditions associated with sub-Alfvénic solar wind.

Fig. 2 (panels f1 and g1) shows the 2D electron velocity distribution functions (eVDFs) observed within the two magnetic clouds. The eVDFs are shown in the $v_{||} - v_{\perp 2}$ plane, where $||$ is along the magnetic field (B) and $v_{\perp 2}$ is in the direction of $\mathbf{v}_e \times \mathbf{B}$, where $\mathbf{v}_e$ is the electron velocity vector and $\mathbf{B}$ is magnetic field vector. These 2D distributions are energy weighted ($\propto v^4 f_e$) that has been demonstrated to effectively resolve distinct magnetic topologies associated with the Alfvén wing transformation (H. Gurram et al. 2025). We use these energy flux densities primarily to characterize

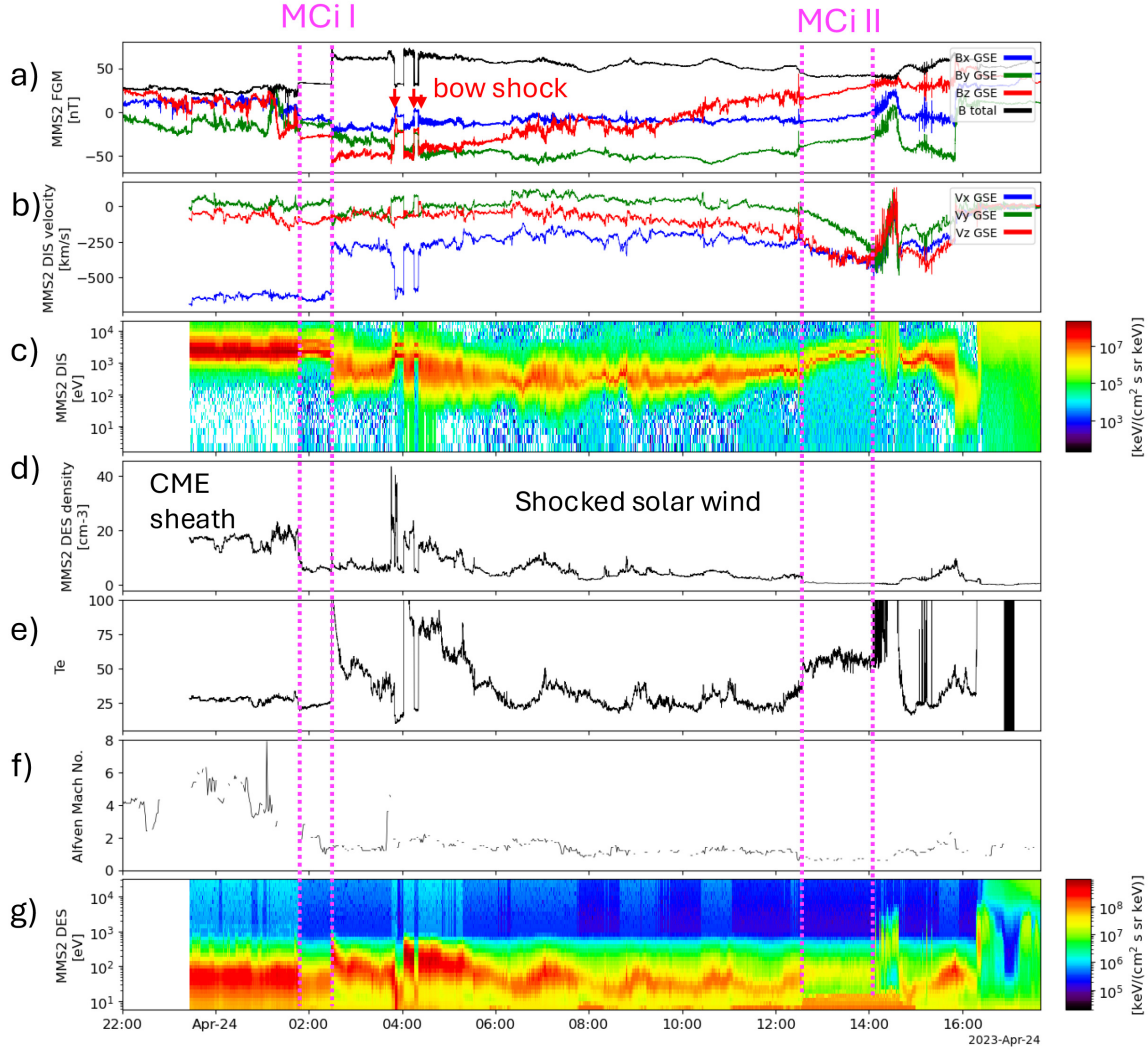


Figure 1. MMS 2 observations of sheath, super-Alfvénic MC (marked MCi-I) and sub-Alfvénic MC (marked MCi-II) driven by the CME on 2023-04-23. a) Magnetic field, b) ion velocity, c) omni-directional ion energy flux, d) electron density, e) electron temperature, f) Alfvén Mach number (from OMNI) and g) omni-directional electron energy flux.

electron populations within the two MCs and CME sheath region. Both MCi-I (Fig. 2f1) and MCi-II (Fig. 2g1) exhibit a narrow beam in the anti-sunward direction ($v_{\parallel} < 0$) which corresponds to strahl electrons, having an energy of 272 eV at 1 AU (J. E. Borovsky 2021). However, the strahl in MCi-II is noticeably less dense than that observed in MCi-I. The eVDF within MCi-II exhibits significant depletion in the electron population (within the 50 eV energy contour) relative to MCi-I. Furthermore, the distribution in MCi-II is more isotropic and Maxwellian-like in the 50–100 eV energy range, whereas MCi-I displays pronounced anisotropy. The electron distributions in MCi-II also differ significantly from those in the CME sheath region. In particular, electrons within MCi-II are substantially hotter, with temperatures of approximately ~ 60 eV, compared to those in both the CME sheath and MCi-I.

This elevated temperature (Fig. 1e) is comparable to typical suprathermal electron energies observed at 1 AU (Š. Štverák et al. 2009; L. Berčič et al. 2020), suggesting that the electron population within MCi-II is predominantly composed of suprathermal electrons.

The 1D cuts of the electron VDFs (f_e , not energy-weighted) inside MCi-II (Fig. 2f2) reveal pronounced suprathermal tails in both MCs in the anti- $\parallel$ direction. In particular, the 1D cuts (Fig. 2f2 and g2) show a clear depletion in the electron population within MCi-II in the ~ 15 –50 eV range compared to MCi-I. Differences are also evident in the strahl component between 50–284 eV: the counts (or decay rate) inside MCi-I decrease more rapidly than in MCi-II, indicating a stronger suprathermal electron population in MCi-II within this energy range. These observations suggest

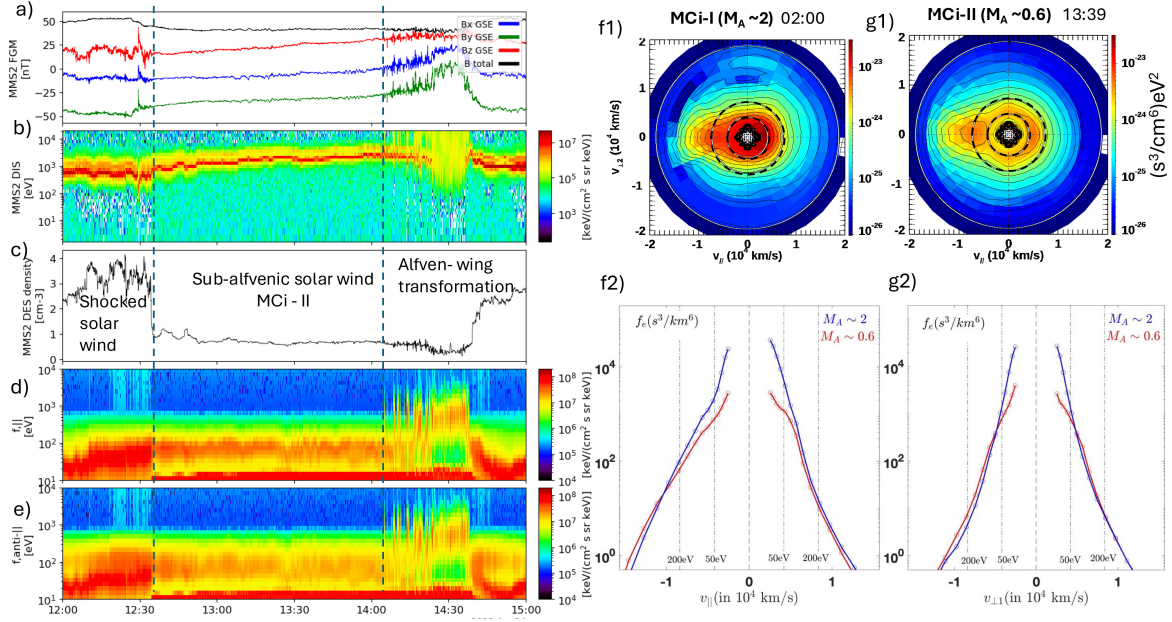


Figure 2. (a) Magnetic field components, (b) omni-directional ion energy flux, (c) electron density, (d) parallel and (e) anti-parallel electron energy fluxes highlighting the sub-Alfvénic MC (MCI-II). Panels (f1) and (g1) show electron energy velocity distribution functions (VDFs) inside the super-Alfvénic (MCI-I) and sub-Alfvénic (MCI-II) MCs, respectively. Dashed circles in the eVDFs mark the energy levels at spacecraft potential ($V_{s/c}$), 50eV and 150eV. Panels (f2) and (g2) show 1D cuts of electron VDFs along the $\parallel$ and $\perp$ 1 directions respectively inside MCI-I and MCI-II. Low-energy photoelectron contamination is removed from all VDFs, which leads to a lack of electron counts at velocities below $|v| \leq 2000 \text{ km/s}$ (D. J. Gershman et al. 2017).

that the elevated electron temperatures observed inside the sub-Alfvénic solar wind interval (Fig. 1e) are primarily due to the enhanced superthermal population which have coronal origin, along with the depletion between 15–50 eV. The uncertainties in FPI measurements at each data point lie in the range of 5–20% in both MCI-I and MCI-II, within which MCI-I remains distinguishable from MCI-II.

MMS observed a sheath region driven by the CME from 23:30UT to $\sim 01:50$ UT (as shown in Fig. 3). The CME sheath has a density about $\sim 15 \text{ cm}^{-3}$ with magnetic field $|B| \sim 30 \text{ nT}$, marked by the yellow interval in Fig. 3. Fig. 3d) and e) shows the parallel and anti-parallel energy flux of electrons, where parallel refers to parallel to B . The CME sheath exhibits electron populations with energy of $\sim 200 \text{ eV}$ in the anti-parallel direction (Fig. 3e), that form a narrow beam (in $v_{\parallel} < 0$) in the eVDF as shown in Fig. 3 f1). The narrow beam corresponds to strahl electrons, which have the mean energy of 272eV at 1 AU (J. E. Borovsky 2021). Inside the CME sheath, there are isolated regions with enhancements of parallel energy fluxes up to $\sim 1 \text{ keV}$ highlighted by the black bar in Fig. 3d). These isolated regions range from few seconds to about 15 minutes. Fig. 3 f2) shows the electron VDF

inside these isolated energetic regions. The parallel energy flux and the distribution looks identical in the anti-parallel direction ($v_{\parallel} < 0$), however along the field aligned direction one sees a broader beam of energetic electrons extending upto $\sim 20,000 \text{ km/s}$ in the velocity space. Unlike the narrow strahl, these energetic electrons show a broader beam almost like concentric circles in $v_{\parallel} > 0$. For electrons inside MCI-I and CME sheath region one major and noteworthy difference is the shape of the eVDF of the core electrons population. Inside the CME sheath the core electrons are denser, hotter ($\sim 40 \text{ eV}$) and isotropic, contrastingly the electrons inside the MCI-I are less dense, colder ($\sim 20 \text{ eV}$) and anisotropic. The eVDF inside the MCI-I exhibits strong core and strahl components, and the core population is slightly anisotropic.

In our recent work (H. Gurram et al. 2025), we have characterized electrons in the magnetic cloud driven by this CME, wherein we have shown that the presence of strahl electrons in the eVDF shows the magnetic connectivity to the solar corona. And energetic electrons of $\sim 1 \text{ keV}$ were seen on the field lines corresponding to dawn and dusk Alfvén wing later during this event, however they originate due to the reconnection between the Earth’s closed field line (CFLs) and the draped IMF

	t_0	t_f	M_A	B [nT]	β	V_A [km/s]	V_i [km/s]	T_i [eV]	T_e [eV]	d_i [km]	d_e [km]
MCi-I	01:50	02:29	2	32.5	0.3	325	650	120	21	102	2.4
MCi-II	12:40	14:03	0.6	41	0.05	950	510	200	58	243	5.6

Table 1. Plasma parameters of the CME magnetic cloud during super-Alfvénic (MCi-I) and sub-Alfvénic (MCi-II) solar wind intervals. (t_0, t_f) specify the beginning and end of the time interval; M_A is the Alfvén Mach number; B is the magnetic field strength; β is the plasma beta; V_A is the Alfvén speed; V_i is the bulk ion velocity; T_i and T_e are the ion and electron temperatures, respectively; and d_i and d_e are the ion and electron inertial lengths.

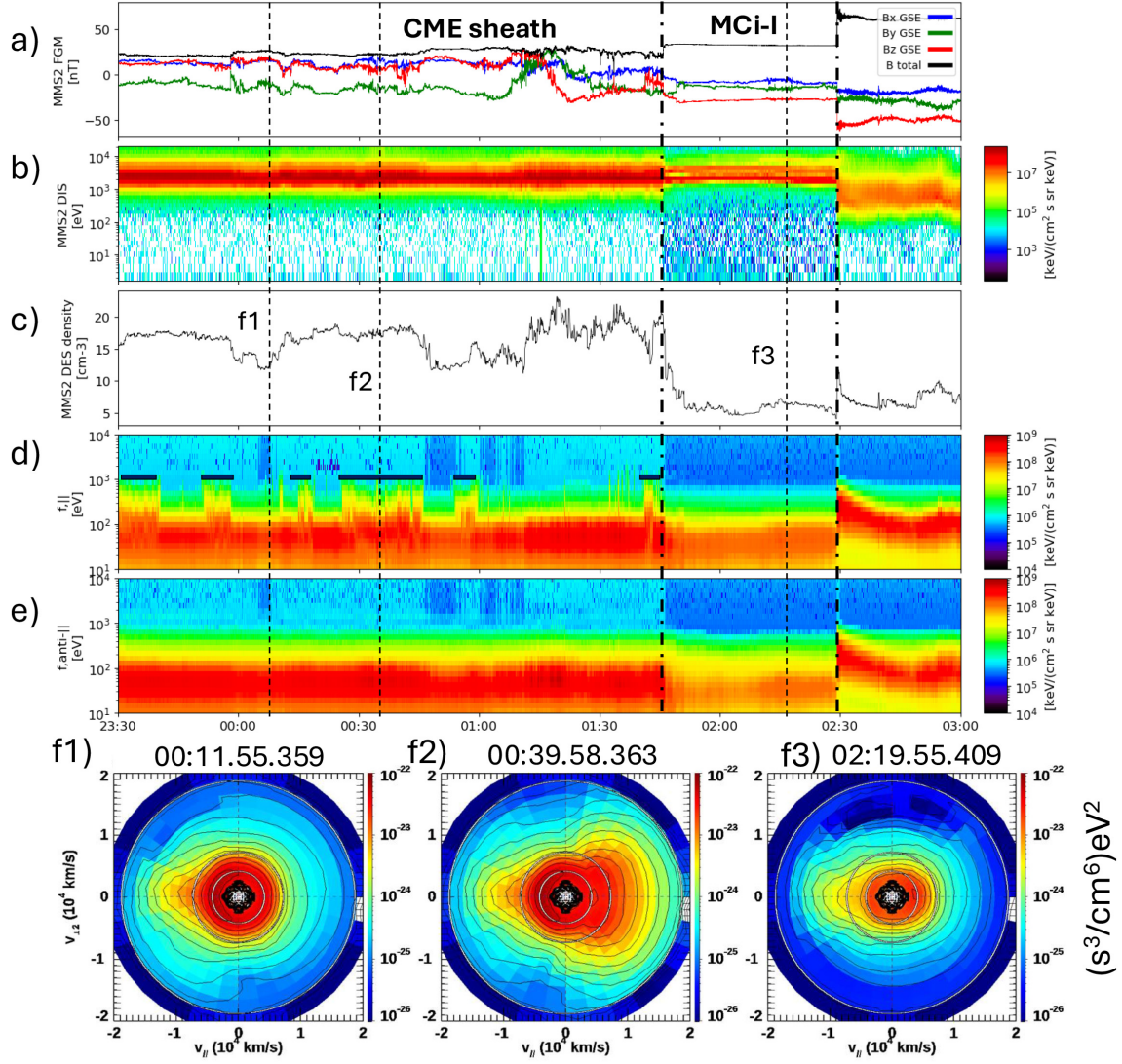


Figure 3. a) Magnetic field, b) omni-directional ion energy flux, c) electron density, e) parallel and f) anti-parallel electron energy fluxes inside CME sheath and super-Alfvénic MC. Solar wind electrons VDFs inside f1) CME sheath, f2) CME sheath with localized energetic electrons and f3) super-Alfvénic MC (MCi-I).

fieldline and these electrons were observed parallel(anti-parallel) to the field line for dusk(dawn) wing. The eVDF in Fig. 3f1) shows the strahl electrons in the CME sheath suggesting that the one footpoint corresponding to this fieldline (as these electrons are magnetized and con-

nected to the fieldline) is on the solar corona. The eVDF in Fig. 3f2) suggests that one foot point is connected to the solar corona, and our potential explanation for energetic electrons along the field line is: these are strahl electrons in parallel directions, and might correspond to

CFL regions in the CME sheath. These CFLs have foot-points on the regions of the solar corona that produce highly energetic electrons. The lack of a narrow beam in the field-aligned strahl component, together with the presence of resonant electrons, suggests that the field-aligned strahl electrons may have undergone scattering by whistler waves generated below the coronal base (C. Vocks & G. Mann 2003). The observed beam asymmetry could also result from wave-particle interactions occurring in the solar wind or within the MC itself during its propagation to Earth.

3. TURBULENCE PROPERTIES OF SUB-ALFVÉNIC SOLAR WIND AT 1AU

This section investigates turbulence within magnetic clouds, highlighting its comparative features relative to turbulence in CME sheaths. We employ spectral slope analysis as the key diagnostic tool to investigate the nature of fluctuations in turbulent systems across multiple scales, which captures key processes such as energy injection or driving, cascade dynamics, and kinetic dissipation (T. S. Horbury et al. 2008; R. Bandyopadhyay et al. 2021; T. N. Parashar et al. 2018). The omnidirectional magnetic field spectrum is calculated using fast Fourier transforms (FFT) of three orthogonal components and summing the power of these components. This gives magnetic spectra as a function of the spacecraft-frame frequency and can be converted to wavenumber using Taylor Hypothesis (G. I. Taylor 1938; J. R. Jokipii 1973). For both the magnetic cloud intervals omnidirectional magnetic field spectra was computed for the low-frequency (< 32 Hz) utilizing FGM (C. T. Russell et al. 2016), as shown in Fig.2a) and b).

The average magnetic field and ion bulk flow vectors for the super- and sub-Alfvénic intervals, in GSE coordinates, are $\mathbf{B}_{\text{MCi-I}} = (-8, -14, -28)$ nT, $\mathbf{V}_{\text{MCi-I}} = (-642, 22, -105)$ km/s, and $\mathbf{B}_{\text{MCi-II}} = (-6.7, -33, 23)$ nT, $\mathbf{V}_{\text{MCi-II}} = (-349, -154, -338)$ km/s, respectively. The angle between the magnetic field and the bulk ion flow, θ_{BV} , is $\sim 69^\circ$ during the super-Alfvénic and $\sim 91^\circ$ sub-Alfvénic interval, indicating that fluctuations in the latter are predominantly perpendicular to the magnetic field ($k_\perp$). It is important to note that for sub-Alfvénic solar wind intervals, power spectra cannot be converted to wavenumber space using the Taylor hypothesis, $k = f/\langle V \rangle_i$. This is in contrast to the super-Alfvénic solar wind, where the condition $V_{\text{SW}} (= V_i) \gg V_A$ is typically satisfied. Within the magnetic cloud MC-II, however, the solar wind speed $V_{\text{SW}} \ll V_A$ does not strictly meet this criterion, rendering the Taylor hypothesis inapplicable (W. H. Matthaeus & M. L.

Goldstein 1982; S. Perri & A. Balogh 2010). However, one can use the assumption that the spacecraft velocity in the plasma frame is “sufficiently oblique” and connect spacecraft frequencies to the field-perpendicular wavevector $k_\perp$ i.e. $\tan \theta_{\text{BV}} \gg k_\parallel/k_\perp \sim \delta u_0/V_A$, where δu_0 is the fluctuations in velocity (~ 33 km/s) for sub-Alfvénic interval (J. C. Perez et al. 2021). For the super-Alfvénic interval we use the Taylor hypothesis and convert the frequency space into wavenumber.

Comparison of the magnetic fluctuations and power spectra across the MCi-I, and MCi-II intervals reveals several key features: i) $\delta B/B$ for both intervals is ~ 0.01 ii) at the injection scale, the sub-Alfvénic interval shows magnetic fluctuation power approximately three times higher than the super-Alfvénic case which is consistent with the enhanced magnetic field inside the MCi-II, iii) in the inertial range, the super-Alfvénic interval displays a shallower spectral slope of -1.4 than the canonical $-5/3$ whereas the sub-Alfvénic interval exhibits a steeper slope of -2 , iv) the magnetic power spectrum in the sub-Alfvénic interval does not show a clear spectral break at ion or sub-ion scales and v) the super-Alfvénic interval exhibits a spectral break near ~ 1 Hz, consistent with ion kinetic scales at 1 AU. A spectral slope of $-5/3$ in the inertial range, corresponding to Kolmogorov scaling, and a slope of -2.8 in the kinetic range are widely regarded as universal features of turbulence in both fast and slow solar wind streams (O. Alexandrova et al. 2009; R. Bruno et al. 2017). These slopes are indicative of a well-developed turbulent cascade from fluid to kinetic scales. However, the deviations are observed inside, sub-Alfvénic MCi-II pointing to the result of weak-MHD turbulence in the low-frequency regime. The absence of a clear spectral break and the steeper slope of -2 at all scales suggest a faster cascade process at all scales inside MCi-II. The magnetic spectral slope of slope -2 in the $k_\perp$ space with $\delta B/B$, and correlation lengths $l_\parallel \sim 1795d_i$ and $l_\perp \sim 6390d_i$ gives $\tau_A/\tau_{nl} = (l_\parallel/v_A)(\delta v/l_\perp) \sim 0.01$. The small scale fluctuations inside MCi-II with a strong background B are all indicative of the presence of weak MHD-turbulence (S. Galtier et al. 2000; C.-S. Ng & A. Bhattacharjee 1996) often seen in Jovian magnetospheres (J. Saur et al. 2002; J. Saur 2021).

Weak MHD turbulence consists of Alfvén waves propagating along magnetic field lines in both directions, interacting weakly by distorting one another without significantly altering their individual energies. These waves must undergo many such interactions before appreciable energy transfer occurs. In contrast,

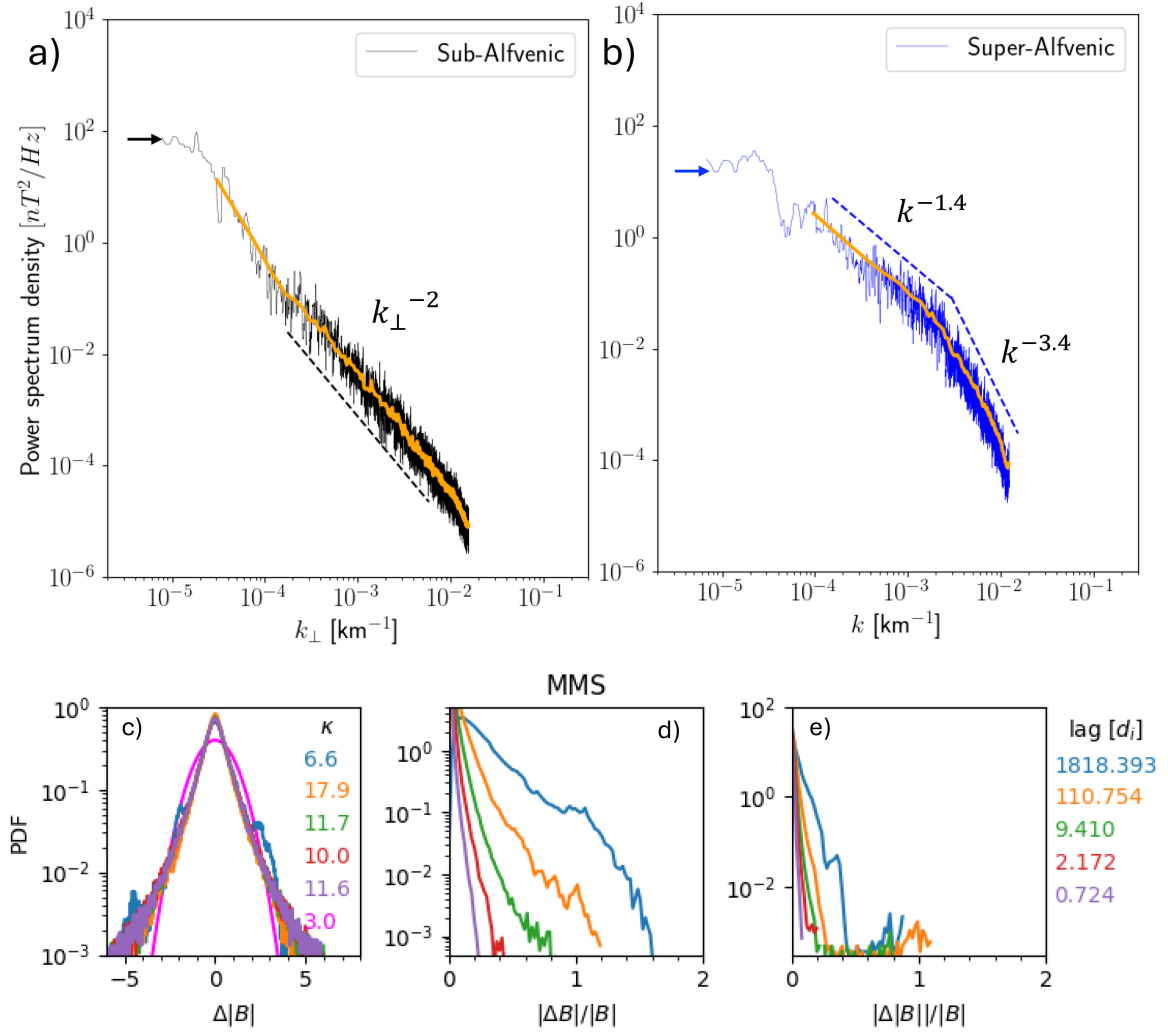


Figure 4. Omnidirectional power spectrum of the magnetic field magnitude B within (a) the sub-Alfvénic (MCi-II) and (b) super-Alfvénic (MCi-I) magnetic cloud intervals, whose plasma parameters are listed in Table 1. (c) Standard normal increments of $|B|$ indicate that intermittency peaks at ion kinetic scales (orange). (d) Normalized increments demonstrate a progressive transition toward Gaussian statistics as the scale size decreases. (e) Magnetic compressibility is very small and becomes both smaller and more Gaussian with decreasing scale.

strong MHD turbulence is characterized by nonlinear interactions that lead to energy cascade on the same timescale as the wave collisions themselves leading to an imbalanced turbulence where more energy is stored in one direction (Y. Lithwick & P. Goldreich 2003). Inside MCi-II, the nonlinear interactions inhibit the parallel cascade due to the wave resonance condition ($\omega = \pm kv_A$), which restricts energy transfer to larger perpendicular wavenumbers only. This leads to an anisotropic cascade of energy in wavevector space resulting in an imbalanced MHD turbulence (S. Sridhar & P. Goldreich 1994; P. Goldreich & S. Sridhar 1997). But, inside super-Alfvénic MCi, the turbulence occurs at scales immediately below the injection scale, where the turbulence cascade is hydrodynamic in nature. For scales below the Alfvén scale, the magnetic field is am-

plified and dominates locally to yield weak turbulence (O. Fornieri et al. 2021). The resulting MHD turbulence is isotropic and the energy cascade resembles hydrodynamic turbulence, following the classic Kolmogorov scaling with exhibits a spectral break. Furthermore, in MCi-II, the alignment between the velocity and the magnetic field is almost perpendicular resulting in a negligible cross helicity and facilitating strong nonlinear interactions. These strong nonlinear interactions lead to a more rapid transfer of energy to smaller scales and enhanced dissipation within MCi-II, thereby producing a steeper slope in the resulting turbulence spectrum. However, for MCi-I, the angle between the velocity and magnetic field is approximately 70° , therefore the resulting cross helicity is relatively larger. This reduces the nonlinear interactions (S. Servidio et al. 2008),

which in turn suppresses the energy transfer to small scales and slows down the energy cascade, resulting in a shallow energy spectrum inside MCi-I.

We also investigated the intermittency of magnetic fluctuations inside the MCi-II to verify if these fluctuations corresponding to weak MHD turbulence exhibit strong intermittent structures which is typical of the solar wind as shown in several studies [Bur \(1991\)](#); [E. Marsch & C. Y. Tu \(1994\)](#); [C. Pagel & A. Balogh \(2002\)](#); [E. Yordanova et al. \(2009\)](#). Intermittency describes the inhomogeneity in the energy transfer between scales and is manifested as a lack of self-similarity in fluctuation distributions between scales arising from coherent structures, such as current sheets and discontinuities ([E. K. J. Kilpua et al. 2020](#)). We compute the intermittency of magnetic fluctuations by computing the probability density functions (PDFs) of the increments. This involves calculating and standardizing the increments at various lags, constructing histograms to generate the PDFs, computing kurtosis, and returning the resulting distributions along with their bin centers and corresponding lag values ([M. R. Argall et al. 2025](#); [S. W. Good et al. 2020](#)). Fig. 4c-e) shows the intermittency of all the three magnetic field components determined from the PDF of $(\Delta B_i^l(t) - \mu)/\sigma$, where μ and σ are the mean and standard deviation of $\Delta B_i^l(t)$, respectively. Here, $\Delta B_i^l(t)$ are increments of the magnetic field computed for a single spacecraft as $\Delta B_i^l(t) = B(t+\tau) - B(t)$, where $l = \langle V_i \rangle \tau$ is the spatial lag determined from temporal lag τ assuming the Taylor approximation ([M. R. Argall et al. 2025](#)). At the inertial scales the PDFs have a kurtosis, $\kappa = 3$ indicating a sub-Gaussian PDF. The PDFs develop intermittent tails as scale size decreases and become super-Gaussian, $\kappa > 3$, exhibiting signs of intermittent turbulence at ion scales. The magnetic fluctuations inside the sub-Alfvénic MCi exhibit less intermittency compared to the turbulent CME sheath which show a kurtosis value of $\kappa \sim 5$ at ion and sub-ion scales shown in ([M. R. Argall et al. 2025](#)). Fig. 4e) shows the magnetic compressibility $|\Delta B|/|B|$ is very small and becomes smaller and more Gaussian as the scale decreases for MCi-II. This further confirms that magnetic fluctuations are incompressible at all scales and predominantly Alfvénic. Furthermore, with $\delta n / \langle n \rangle \approx 0.07/0.7 = 0.1$, the observed density variations are very small, indicating the sub-Alfvénic region is largely incompressible.

4. CONCLUSIONS AND DISCUSSIONS

In this paper, utilizing MMS observations, we conducted a comparative study of sub-Alfvénic and super-Alfvénic MCis, both comprising low-beta, mag-

netically dominated coronal plasma at 1 AU. Typically, solar wind conditions farther from the Sun are super-Alfvénic due to expansion through interplanetary space, and sub-Alfvénic solar wind near Earth is a rare occurrence. However, a unique opportunity arose during the 2023 CME event, which allowed us to observe sub-Alfvénic solar wind in the near-Earth environment. This rare event provided valuable insights into the behavior and properties of solar wind under sub-Alfvénic conditions.

We characterized the electron VDFs inside the CME sheath, sub-and super Alfvénic MC's. The electrons inside all these three regions show a anti-field aligned beam i.e. strahl electrons. The electrons inside the CME sheath show isolated regions of heating wherein the eVDFs show a energetic ~ 1 keV electrons along the magnetic field. We also found that electrons inside the sub-Alfvénic MCi are significantly hotter than those in the CME sheath and the super-Alfvénic MCi. The sub-Alfvénic solar wind is characterized by low-density, low-beta plasma ($\beta \approx 0.03$). Typically, solar wind electron velocity distribution functions (VDFs) consist of a thermal core and superthermal halo and strahl components, with the core contributing approximately 80–90% of the total electron density. The FPI measurements cannot accurately capture the core thermal electrons (< 20 eV) in the solar wind; however, they measure the other populations with an uncertainty of about 20%. The electron VDFs observed within the sub-Alfvénic MC show strong superthermal electron populations and a depletion in electrons between 15–50 eV compared to the super-Alfvénic MC, resulting in elevated temperatures inside the sub-Alfvénic MC. The depletion of electron populations below 50 eV in the sub-Alfvénic MCi may be responsible for the observed low-density conditions within the structure. This depletion could be linked to formation conditions in the corona, where whistler waves generated below the coronal base can resonate with electrons, preferentially enhancing the high-energy population and producing superthermal tails while leaving the core underpopulated ([C. Vocks & G. Mann 2003](#); [C. Vocks et al. 2008](#); [C. Vocks 2021](#)).

We conducted the first analysis of turbulence properties inside the sub-Alfvénic MCi (MCi-II) at 1 AU and compared the properties with the turbulent CME sheath region as well as the super-Alfvénic MC (MCi-I). The fluctuations inside the MCi-I and MCi-II are in the order of ~ 0.01 less than those observed in the CME sheath. The magnetic spectra inside MCi-II exhibited

a power law with steeper than Kolmogorov scaling and had a spectral slope of -2 in the inertial range with no clear spectral break at ion or sub-ion scales. This is different from the turbulence inside the MCi-I and CME sheath which exhibit a power-law scaling of approximately $k^{-5/3}$ a characteristic of MHD turbulence theory, and observed both upstream at L1 and near Earth (M. R. Argall et al. 2025). Other statistical studies on sheath turbulence have shown spectral slopes for magnetic fluctuations in the majority of cases are steeper than $-5/3$. Sheaths had on average steeper inertial-range spectral indices than the Kolmogorov ($k^{-5/3}$) and Kraichnan ($k^{-3/2}$) indices. However, the range of indices was relatively large ($\sim [-2.2, -1.3]$) and was larger for slow sheaths and sheaths preceded by slow solar wind (E. K. J. Kilpua et al. 2020, 2021).

Recent studies using Parker Solar Probe (PSP) have, for the first time, provided detailed observations of sub-Alfvénic solar wind originating from the solar corona, below the Alfvén surface (R. Bandyopadhyay et al. 2022; T. Alberti et al. 2022; J. Zhang et al. 2022; L. L. Zhao et al. 2022; G. P. Zank et al. 2022). G. P. Zank et al. (2022) analyzed the spectral characteristics of this regime and reported strong consistency with predictions from nearly incompressible MHD theory. More recently, T. Alberti et al. (2022) investigated the scaling properties of both sub- and super-Alfvénic solar wind and identified convex scaling exponents in the inertial range and linear scaling at sub-ion scales for both regimes.

In contrast, the sub-Alfvénic interval (MCi-II) at 1 AU shows a steeper magnetic field power-law slope of -2 in the inertial range, suggesting a weak-MHD turbulence cascade. This is similar to sub-ion scale kinetic Alfvén wave turbulence in the low-frequency regime similar to the found in Io’s flux tube (S. Janser et al. 2022). The fluctuations inside MCi-II have negligible cross-helicity, resulting in stronger non-linear interactions and faster energy cascade compared to MCi-I and CME sheath. The magnetic field structures inside the MCi-II show less intermittency compared to CME sheath at the ion scales, but the fluctuations are seen to develop intermittency at the kinetic scales linking to dissipation and current sheets at these scales. The density and magnetic variations are also minimal suggesting the steady sub-Alfvénic solar wind at 1 AU is incompressible.

ACKNOWLEDGMENTS

We gratefully acknowledge the MMS mission team and instrument principal investigators for providing access to high-quality data and for their continued support. We also thank the FPI Visualizer team for their assistance and for making their visualization tools openly accessible at <https://fpi.gsfc.nasa.gov/>. Our appreciation extends to the pySPEDAS development team for offering robust data analysis tools that supported this study. HG acknowledges partial support of NSF grant AGS2247718.

REFERENCES

- 1991, *Journal of Geophysical Research: Space Physics*, 96, 5847, doi: <https://doi.org/10.1029/91JA00087>
- Alberti, T., Benella, S., Carbone, V., et al. 2022, *Universe*, 8, 338, doi: [10.3390/universe8070338](https://doi.org/10.3390/universe8070338)
- Alexandrova, O., Saur, J., Lacombe, C., et al. 2009, *Physical Review Letters*, 103, doi: [10.1103/physrevlett.103.165003](https://doi.org/10.1103/physrevlett.103.165003)
- Argall, M. R., Chen, L.-J., Lugaz, N., et al. 2025, <https://arxiv.org/abs/2505.13645>
- Bandyopadhyay, R., McComas, D. J., Szalay, J. R., et al. 2021, *Geophysical Research Letters*, 48, e2021GL095006, doi: <https://doi.org/10.1029/2021GL095006>
- Bandyopadhyay, R., Matthaeus, W. H., McComas, D. J., et al. 2022, *ApJL*, 926, L1, doi: [10.3847/2041-8213/ac4a5c](https://doi.org/10.3847/2041-8213/ac4a5c)
- Berčić, L., Larson, D., Whittlesey, P., et al. 2020, *The Astrophysical Journal*, 892, 88, doi: [10.3847/1538-4357/ab7b7a](https://doi.org/10.3847/1538-4357/ab7b7a)
- Borovsky, J. E. 2021, *Frontiers in Astronomy and Space Sciences*, 8, doi: [10.3389/fspas.2021.646443](https://doi.org/10.3389/fspas.2021.646443)
- Bowers, C. F., Jackman, C. M., Jia, X., et al. 2025, *Journal of Geophysical Research: Space Physics*, doi: [10.1029/2024JA033619](https://doi.org/10.1029/2024JA033619)
- Bruno, R., Telloni, D., DeIure, D., & Pietropaolo, E. 2017, *MNRAS*, 472, 1052, doi: [10.1093/mnras/stx2008](https://doi.org/10.1093/mnras/stx2008)
- Burch, J. L., Torbert, R. B., Phan, T. D., et al. 2016, *Science*, 352, aaf2939, doi: [10.1126/science.aaf2939](https://doi.org/10.1126/science.aaf2939)
- Burkholder, B. L., Chen, L.-J., Sarantos, M., et al. 2024, *Geophysical Research Letters*, 51, e2024GL108311, doi: [10.1029/2024GL108311](https://doi.org/10.1029/2024GL108311)
- Burlaga, L. F., Sittler, E., Mariani, F., & Schwenn, R. 1981, *Journal of Geophysical Research: Space Physics*, 86, 6673, doi: [10.1029/JA086iA08p06673](https://doi.org/10.1029/JA086iA08p06673)
- Chané, E., van der Holst, B., Jacobs, C., Poedts, S., & Kimpe, D. 2012, *Journal of Geophysical Research: Space Physics*, 117, A09106, doi: [10.1029/2012JA017834](https://doi.org/10.1029/2012JA017834)

- Chen, L.-J., Gershman, D., Burkholder, B., et al. 2024, *Geophysical Research Letters*, 51, e2024GL108894, doi: <https://doi.org/10.1029/2024GL108894>
- Chen, Y., Dong, C., Chen, L.-J., Sarantos, M., & Burkholder, B. L. 2025, *Geophysical Research Letters*, 52, e2024GL113416, doi: <https://doi.org/10.1029/2024GL113416>
- Fornieri, O., Gaggero, D., Cerri, S. S., Vittino, A., & Grasso, D. 2021, *Monthly Notices of the Royal Astronomical Society*, 502, 5821, doi: [10.1093/mnras/stab387](https://doi.org/10.1093/mnras/stab387)
- Galtier, S., Nazarenko, S. V., Newell, A. C., & Pouquet, A. 2000, *Journal of Plasma Physics*, 63, 447, doi: [10.1017/S0022377899008284](https://doi.org/10.1017/S0022377899008284)
- Gershman, D. J., Avanov, L. A., Boardsen, S. A., et al. 2017, *Journal of Geophysical Research: Space Physics*, 122, 11,548, doi: <https://doi.org/10.1002/2017JA024518>
- Goldreich, P., & Sridhar, S. 1997, *The Astrophysical Journal*, 485, 680, doi: [10.1086/304442](https://doi.org/10.1086/304442)
- Good, S. W., Ala-Lahti, M., Palmerio, E., Kilpua, E. K. J., & Osmane, A. 2020, *The Astrophysical Journal*, 893, 110, doi: [10.3847/1538-4357/ab7fa2](https://doi.org/10.3847/1538-4357/ab7fa2)
- Gosling, J. T. 1993, *J. Geophys. Res.*, 98, 18937, doi: [10.1029/93JA01896](https://doi.org/10.1029/93JA01896)
- Gurram, H., Shuster, J. R., Chen, L.-J., et al. 2025, *Geophysical Research Letters*, 52, e2024GL111931, doi: <https://doi.org/10.1029/2024GL111931>
- Hajra, R., & Tsurutani, B. T. 2022, *The Astrophysical Journal*, 926, 186, doi: [10.3847/1538-4357/ac4471](https://doi.org/10.3847/1538-4357/ac4471)
- Horbury, T. S., Forman, M., & Oughton, S. 2008, *PhRvL*, 101, 175005, doi: [10.1103/PhysRevLett.101.175005](https://doi.org/10.1103/PhysRevLett.101.175005)
- Janser, S., Saur, J., Clark, G., Sulaiman, A. H., & Szalay, J. R. 2022, *Journal of Geophysical Research: Space Physics*, 127, e2022JA030675, doi: <https://doi.org/10.1029/2022JA030675>
- Jiao, Y., Liu, Y. D., Ran, H., & Cheng, W. 2023, *The Astrophysical Journal*, 960, 42, doi: [10.3847/1538-4357/ad0dfe](https://doi.org/10.3847/1538-4357/ad0dfe)
- Jokipii, J. R. 1973, *ARA&A*, 11, 1, doi: [10.1146/annurev.aa.11.090173.000245](https://doi.org/10.1146/annurev.aa.11.090173.000245)
- Kilpua, E. K. J., Good, S. W., Ala-Lahti, M., et al. 2021, *Frontiers in Astronomy and Space Sciences*, 7, 109, doi: [10.3389/fspas.2020.610278](https://doi.org/10.3389/fspas.2020.610278)
- Kilpua, E. K. J., Koskinen, H. E. J., & Pulkkinen, T. I. 2017, *Living Reviews in Solar Physics*, 14, 5, doi: [10.1007/s41116-017-0009-6](https://doi.org/10.1007/s41116-017-0009-6)
- Kilpua, E. K. J., Fontaine, D., Good, S. W., et al. 2020, *Annales Geophysicae*, 38, 999, doi: [10.5194/angeo-38-999-2020](https://doi.org/10.5194/angeo-38-999-2020)
- Klein, L. W., & Burlaga, L. F. 1982, *Journal of Geophysical Research: Space Physics*, 87, 613, doi: [10.1029/JA087iA02p00613](https://doi.org/10.1029/JA087iA02p00613)
- Lavraud, B., & Borovsky, J. E. 2008, *Journal of Geophysical Research: Space Physics*, 113, doi: <https://doi.org/10.1029/2008JA013192>
- Lithwick, Y., & Goldreich, P. 2003, *The Astrophysical Journal*, 582, 1220, doi: [10.1086/344676](https://doi.org/10.1086/344676)
- Marsch, E., & Tu, C. Y. 1994, *Annales Geophysicae*, 12, 1127, doi: [10.1007/s00585-994-1127-8](https://doi.org/10.1007/s00585-994-1127-8)
- Matthaeus, W. H., & Goldstein, M. L. 1982, *Journal of Geophysical Research: Space Physics*, 87, 6011, doi: <https://doi.org/10.1029/JA087iA08p06011>
- Ng, C.-S., & Bhattacharjee, A. 1996, *The Astrophysical Journal*, 465, 845, doi: [10.1086/177472](https://doi.org/10.1086/177472)
- Pagel, C., & Balogh, A. 2002, *Journal of Geophysical Research: Space Physics*, 107, SSH 6, doi: <https://doi.org/10.1029/2002JA009331>
- Parashar, T. N., Chasapis, A., Bandyopadhyay, R., et al. 2018, *PhRvL*, 121, 265101, doi: [10.1103/PhysRevLett.121.265101](https://doi.org/10.1103/PhysRevLett.121.265101)
- Perez, J. C., Bourouaine, S., Chen, C. H. K., & Raouafi, N. E. 2021, *A&A*, 650, A22, doi: [10.1051/0004-6361/202039879](https://doi.org/10.1051/0004-6361/202039879)
- Perri, S., & Balogh, A. 2010, *ApJ*, 714, 937, doi: [10.1088/0004-637X/714/1/937](https://doi.org/10.1088/0004-637X/714/1/937)
- Pollock, C., Moore, T., Jacques, A., et al. 2016, *Space Science Reviews*, 199, 331, doi: [10.1007/s11214-016-0245-4](https://doi.org/10.1007/s11214-016-0245-4)
- Russell, C. T., Anderson, B. J., Baumjohann, W., et al. 2016, *SSRv*, 199, 189, doi: [10.1007/s11214-014-0057-3](https://doi.org/10.1007/s11214-014-0057-3)
- Saur, J. 2021, *Frontiers in Astronomy and Space Sciences*, 8, 56, doi: [10.3389/fspas.2021.624602](https://doi.org/10.3389/fspas.2021.624602)
- Saur, J., Politano, H., Pouquet, A., & Matthaeus, W. H. 2002, *A&A*, 386, 699, doi: [10.1051/0004-6361:20020305](https://doi.org/10.1051/0004-6361:20020305)
- Servidio, S., Matthaeus, W. H., & Dmitruk, P. 2008, *Physical Review Letters*, 100, 095005, doi: [10.1103/PhysRevLett.100.095005](https://doi.org/10.1103/PhysRevLett.100.095005)
- Sridhar, S., & Goldreich, P. 1994, *ApJ*, 432, 612, doi: [10.1086/174600](https://doi.org/10.1086/174600)
- Taylor, G. I. 1938, *Proceedings of the Royal Society of London Series A*, 164, 476, doi: [10.1098/rspa.1938.0032](https://doi.org/10.1098/rspa.1938.0032)
- Vocks, C. 2021, in *Astrophysics and Space Science Library*, Vol. 464, Kappa Distributions; From Observational Evidences via Controversial Predictions to a Consistent Theory of Nonequilibrium Plasmas, ed. M. Lazar & H. Fichtner, 125–143, doi: [10.1007/978-3-030-82623-9_7](https://doi.org/10.1007/978-3-030-82623-9_7)
- Vocks, C., & Mann, G. 2003, *ApJ*, 593, 1134, doi: [10.1086/376682](https://doi.org/10.1086/376682)

694 Vocks, C., Mann, G., & Rausche, G. 2008, A&A, 480, 527,
695 doi: [10.1051/0004-6361:20078826](https://doi.org/10.1051/0004-6361:20078826)
696 Štverák, Š., Maksimovic, M., Trávníček, P. M., et al. 2009,
697 Journal of Geophysical Research (Space Physics), 114,
698 A05104, doi: [10.1029/2008JA013883](https://doi.org/10.1029/2008JA013883)
699 Webb, D. F., & Howard, T. A. 2012, Living Reviews in
700 Solar Physics, 9, 3, doi: [10.12942/lrsp-2012-3](https://doi.org/10.12942/lrsp-2012-3)

701 Yordanova, E., Balogh, A., Noullez, A., & von Steiger, R.
702 2009, Journal of Geophysical Research: Space Physics,
703 114, doi: <https://doi.org/10.1029/2009JA014067>
704 Zank, G. P., Zhao, L. L., Adhikari, L., et al. 2022, ApJL,
705 926, L16, doi: [10.3847/2041-8213/ac51da](https://doi.org/10.3847/2041-8213/ac51da)
706 Zhang, J., Huang, S. Y., Yuan, Z. G., et al. 2022, ApJ, 937,
707 70, doi: [10.3847/1538-4357/ac8c34](https://doi.org/10.3847/1538-4357/ac8c34)
708 Zhao, L. L., Zank, G. P., Adhikari, L., et al. 2022, ApJL,
709 934, L36, doi: [10.3847/2041-8213/ac8353](https://doi.org/10.3847/2041-8213/ac8353)